

Assignment 1

Explore following 4 websites and answer the questions

- bitcoin.org
- [Blockchain.com](https://blockchain.com/)
- <https://coinmarketcap.com/>
- <https://www.investopedia.com/>

One-Word Answerable Questions from bitcoin.org

Bitcoin Basics

1. Bitcoin was introduced by whom?
->**Satoshi Nakamoto**
2. Bitcoin transactions are recorded on which data structure?
->**Blockchain**
3. Bitcoin uses which consensus mechanism?
->**Proof of Work (PoW)**
4. The smallest unit of Bitcoin is called?
->**Satoshi**
5. Bitcoin is a **peer-to-peer** type of what system?
->**Electronic cash system**

Wallets

6. A wallet stores the user's **Private** key.
7. The recovery phrase used in wallets is called a **Seed (or Mnemonic)** phrase.
8. Hardware wallets store keys **Offline** (online/offline).

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9. Bitcoin addresses are derived from Public keys.
10. What is required to sign a transaction? (public/private)

Security

11. Bitcoin prevents double-spending using Blockchain.
12. Bitcoin is secured using Asymmetric cryptography (asymmetric/symmetric).
13. To run a Bitcoin node, the software required is Bitcoin Core.
14. The process of verifying transactions is called Validation.
15. The Bitcoin network is described as decentralized (centralized/decentralized).

Mining

16. Miners solve Cryptographic puzzles to add blocks.
17. Mining rewards a block with newly created Bitcoins
18. The block time target for Bitcoin is 10 minutes.
19. Mining difficulty adjusts every 2016 blocks
20. Miners compete to find a valid block Hash (hash/signature).

Blockchain Structure

21. Each block contains a Merkle root.
22. Blocks are linked using the previous block's Hash.
23. A Bitcoin transaction contains inputs and Outputs.
24. Unspent transaction outputs are called UTXOs (Unspent Transaction Outputs).
25. The first-ever Bitcoin block is called the Genesis block.

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Network

26. A full node stores the entire blockchain Ledger.
27. Bitcoin uses which protocol for communication? (TCP/P2P)
28. A light wallet uses SPV mode (SPV/full).
29. Nodes relay information across a Peer-to-peer (P2P) network.
30. The maximum Bitcoin supply is 21 million.

Task B1: Explore a Block

Note the following from the **latest block**:

- Block height:**961,124**
- Block
hash:**000000000000000000000004d5d3148f9aef47da584facb242eb3502e3c465d0eab**
- Timestamp:**August 05, 2026, 12:19:23**
- Number of transactions:**4057**
- Miner/pool name:**Unknown**

Task B2: Explore a Transaction

Open any one Bitcoin transaction.

10. Write the TXID of the transaction.
7cf91ed4573bcd285c6b6dfbb576523e65447271dfdf21d496175f678ecf36c7
11. Number of inputs:**1**
12. Number of outputs:**8**
13. Transaction fee:**0 BTC (0 SATS)**
14. How many confirmations does it have:**2**

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Task B3: Explore an Address

15. Search for any Bitcoin address shown in the block:

1K6KoYC69NnafWJ7YgtrpwJxBLiijWqwa6

16. Record: **1K6KoYC69NnafWJ7YgtrpwJxBLiijWqwa6**

- Total BTC received: **0.12772136 BTC**
- Total BTC sent: **0.02073492 BTC**
- Current balance: **0.10698644 BTC**

What is the role of a full node?

A **full node** stores the complete Bitcoin blockchain, independently verifies all transactions and blocks, enforces Bitcoin's protocol rules, and relays valid transactions and blocks across the network, helping keep the Bitcoin network secure and decentralized.

List any two developer resources available on bitcoin.org.

Developer Guides

Reference (API Reference)

Explore : <https://coinmarketcap.com/>

Task C1: Market Information

- Current price of BTC: **\$64,109.14**
- 24-hour % change: **-2.71%**
- Market capitalization: **\$1.27 Trillion**
- Circulating supply: **20,062,306 BTC**
- Total/maximum supply: **21,000,000 BTC**

Task C2: Historical Data

- What was the price of Bitcoin one year ago?
Approximately \$≈ \$115,070 USD
- Note the highest and lowest prices in the last 30 days. **\$66.8K USD**

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Explore <https://www.investopedia.com/>

Why is Bitcoin considered volatile?

Bitcoin is considered **volatile** because its price changes rapidly due to factors such as market demand and supply, investor sentiment, regulatory announcements, macroeconomic events, and speculative trading.

List two investment risks mentioned.

High price volatility

Regulatory uncertainty

♦ Part E: Reflection & Analysis

Which website gave you the most clear understanding? Why?

Bitcoin.org gave me the clearest understanding because it explains Bitcoin concepts in simple and easy-to-understand language. It provides well-organized information about wallets, mining, security, and blockchain, making it suitable for beginners.

In one paragraph, explain how blockchain ensures trust without a central authority.

Blockchain ensures trust without a central authority by storing transactions in a distributed ledger that is shared among thousands of computers (nodes). Every transaction is verified through the network's consensus mechanism before being added to a block. Each block is cryptographically linked to the previous one, making the data secure and difficult to alter. Since every node maintains a copy of the blockchain and independently verifies transactions, no single person or organization controls the network, ensuring transparency, security, and trust.